CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

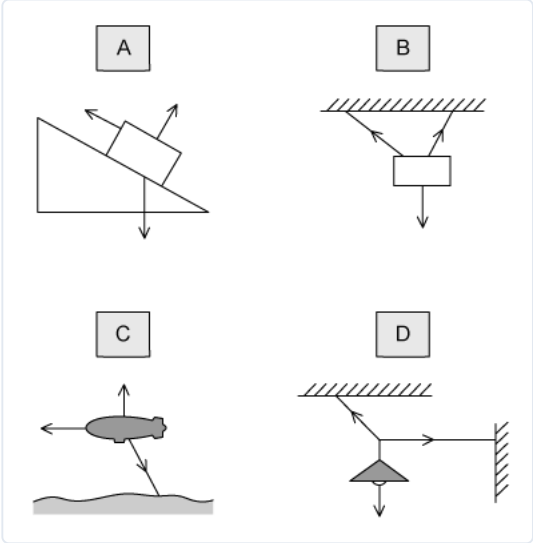
4.2 Equilibrium of Forces

Syllabus Section: 4. Forces, Density & Pressure **Paper Type:** Multiple Choice (Paper 1) **Total:** 23 Questions (23 Marks)

Question 1 Easy [1 mark]

The diagrams show the forces acting on different bodies.

Which body **cannot** be in equilibrium?

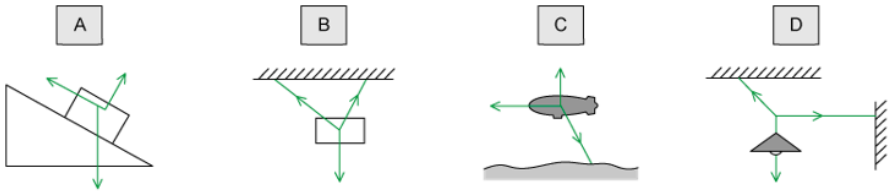


- **A**
- **B**
- **C**
- **D**

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- If the vector arrows form a closed triangle, the resultant force is zero
- However, for a body to be in equilibrium, the resultant moment must also be zero
- Another way to look at it is that three forces can only be in equilibrium if they all pass through the same point
    - If not, then there will be a turning force (resultant moment) on the object
- So, if we join the vector arrows to see where they meet, we can see that the only option where the three forces do not meet at a singular point is option A


A
B
C
D

Question 2 Easy [1 mark]

A force of 5 N pushes a ball due north, and another force of 12 N pushes it due east.

What is the magnitude of the net force acting on the ball?

A 7 N

B 11 N

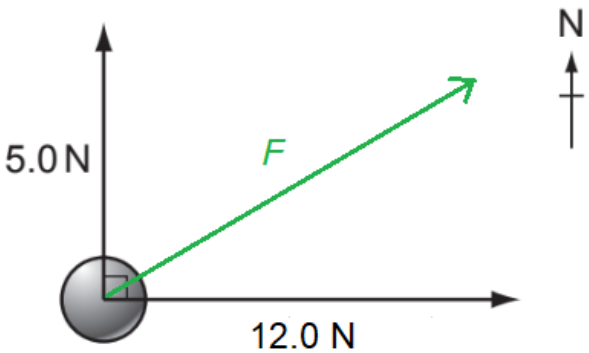
C 13 N

D 17 N

Mark Scheme & Explanation:

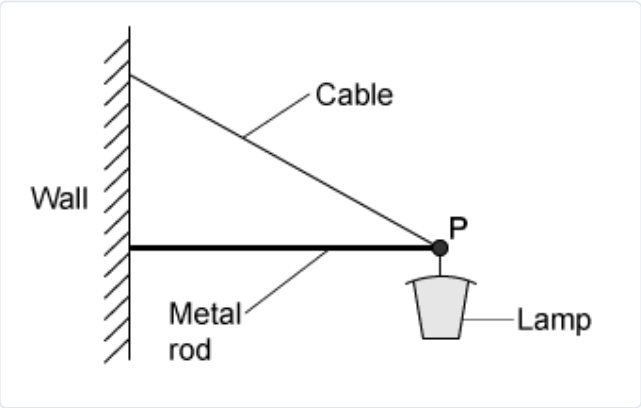
*The correct answer is **C** because:*

- To find the net force for two forces which are perpendicular to each other we can use
  - $F^2 = F_N^2 + F_E^2$
  - $F = \sqrt{5^2 + 12^2} = 13 \text{ N}$

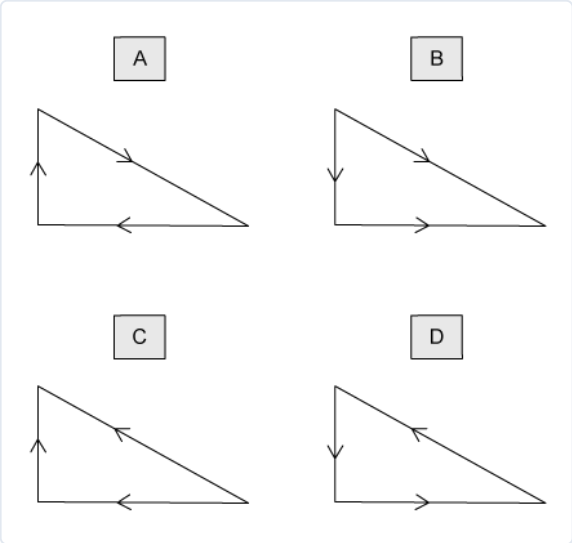


Question 3 Easy [1 mark]

A street lamp is fixed to a wall by a metal rod and a cable.



Which vector triangle represents the forces acting at point P?



A

B

C

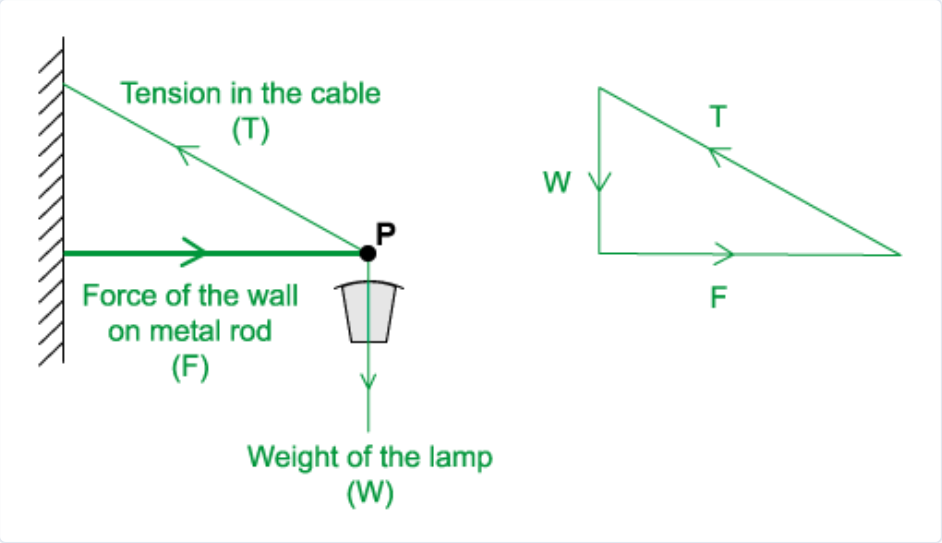
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Weight always acts downwards
- Tension is the pull of the cable on the object so it always acts away from the mass
- Since the lamp is at rest, the three forces on point P must be in equilibrium

- That means they form a closed triangle when drawn tip-to-tail:

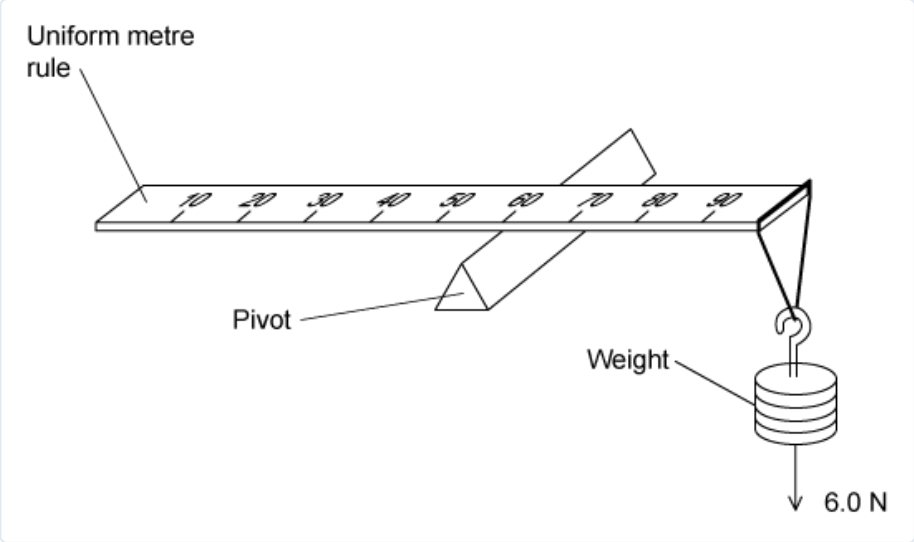


A & C are incorrect because weight always acts downwards but these triangles show the weight acting upwards

B is incorrect because the tension in a wire always acts away from the mass whereas this triangle shows it acting towards the mass

Question 4 Easy [1 mark]

A uniform metre rule of weight 4.0 N is pivoted at the 60 cm mark. A 6.0 N weight is suspended from one end, causing the rule to rotate about the pivot.



At the instant when the rule is horizontal, what is the resultant turning moment about the pivot?

	Resultant moment	Direction
A	2.0 N m	clockwise
B	2.0 N m	anticlockwise
C	2.8 N m	clockwise
D	2.8 N m	anticlockwise

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- To calculate moments, we will need to use
  - $\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$
- We will look at the clockwise and anticlockwise moments separately
- The metre rule is uniform, meaning its weight acts at the centre mark, 50 cm

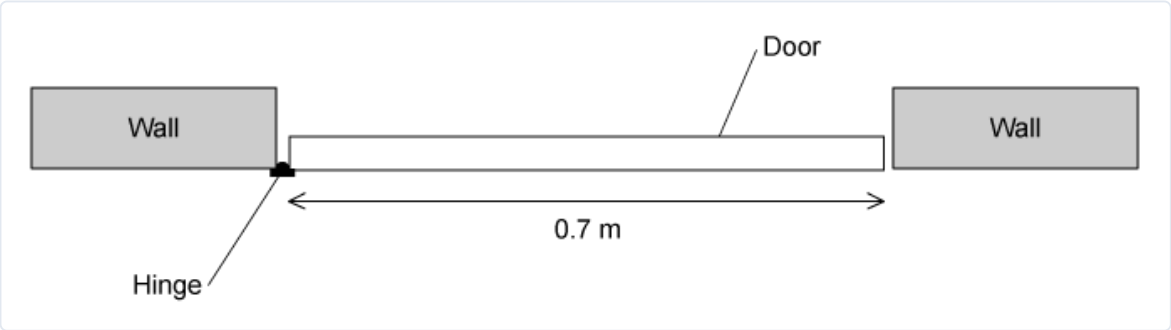
- Therefore, the distance between the pivot and the centre of gravity is 10 cm
- The weight of the ruler produces an anticlockwise moment of
    - 4.0 × 0.1 = 0.4 N m
- The 6.0 N weight acts downwards at a distance of 40 cm from the pivot, producing a clockwise moment of
    - 6.0 × 0.4 = 2.4 N m
- Resultant moment = sum of clockwise and anticlockwise moments
    - 2.4 – 0.4 = 2.0 N m
- Therefore, the resultant turning moment is 2.0 N m clockwise

B & D are incorrect because a positive value is generally taken as a clockwise moment

C is incorrect because this is the answer you get if you add the moments together instead of taking away the anticlockwise moment

Question 5 Easy [1 mark]

The diagram shows a plan view of a door which requires a moment of 14 N m to open it.



What is the minimum force that must be applied at the door's midpoint to ensure it opens?

A 4.9 N

B 9.8 N

C 20 N

D 40 N

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The moment of a force is given by:
  - $\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$
- The midpoint of the door is at 0.35 m
- So minimum force = moment / distance from hinge to midpoint
  - $\frac{14}{0.35} = 40 \text{ N}$

Question 6 Easy [1 mark]

Forces of 3 N, 4 N and 5 N act at one point on an object. The angles at which the forces act can vary.

What is the value of the minimum resultant force of these forces?

A 0 N

B between 0 and 2 N

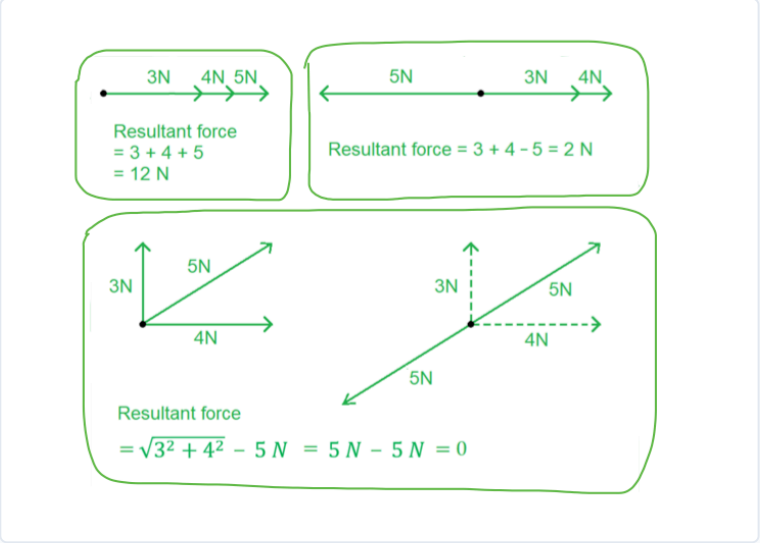
C 2 N

D between 2 N and 4 N

Mark Scheme & Explanation:

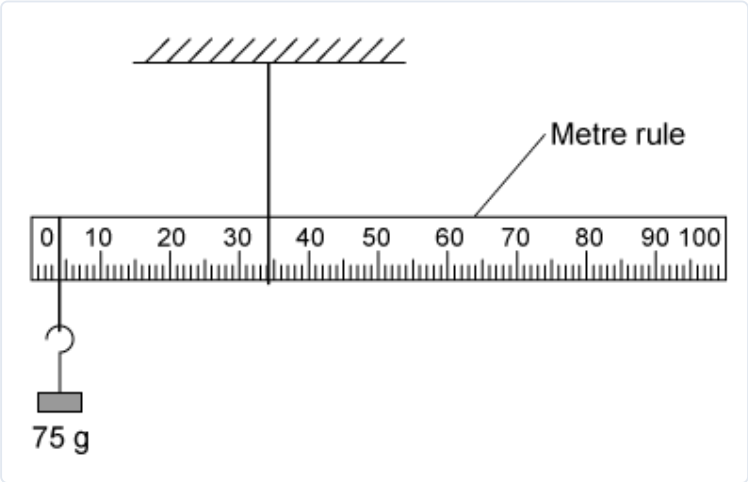
*The correct answer is **A** because:*

- The minimum resultant force is found by finding the combination of forces which produces the smallest value
- Consider the resultant magnitude of 3 N and 4 N when they are perpendicular to each other:
  - Resultant = $\sqrt{3^2 + 4^2} = 5$ N
- So, if the 5 N force is then placed in the opposite direction to the resultant force of the perpendicular 3 N and 4 N forces, the overall resultant force is zero



Question 7 Medium [1 mark]

A uniform metre rule is hung from the ceiling as shown



The rule balances when a 75 g mass is hung from one end of the ruler

What is the mass of the metre rule?

A 38 g

B 51 g

C 141 g

D 159 g

Mark Scheme & Explanation:

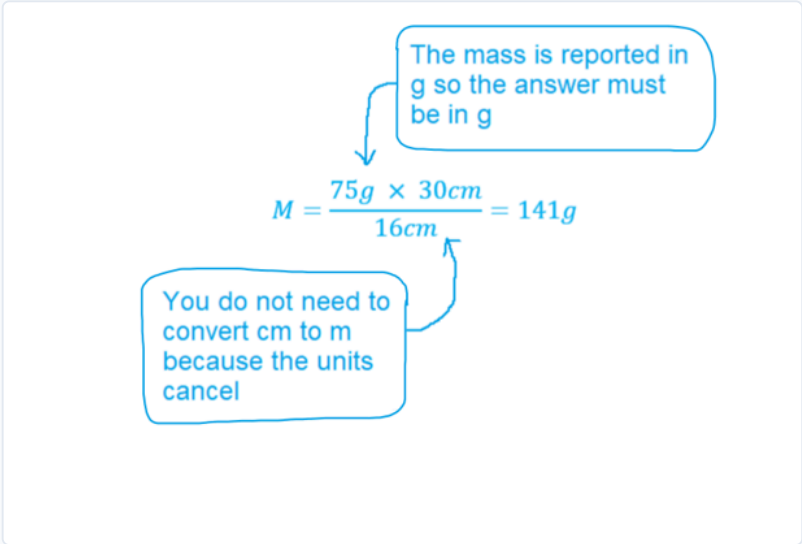
*The correct answer is **C** because:*

- To calculate moments, we will need to use
  - $\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$
- The metre rule is uniform, meaning its weight acts at the centre mark, 50 cm
- The 75g mass acts downwards at the 4 cm marker
- The pivot is positioned at 34 cm, so
  - distance to the centre of gravity = 50 – 34 = 16 cm
  - distance to the 75 g mass = 34 – 4 = 30 cm
- Clockwise moment due to the ruler's weight:

- $Mg \times 16$

- Anticlockwise moment due to the 75 g mass:
  - $75g \times 30$
- For the system to be in equilibrium, clockwise moment = anticlockwise moment
  - $Mg \times 16 \quad = \quad 75g \times 30$
  - $M \quad = \quad \frac{75 \times 30}{16} \quad = \quad 141 \quad \text{g}$
- Therefore, the metre rule has a mass of 141 g

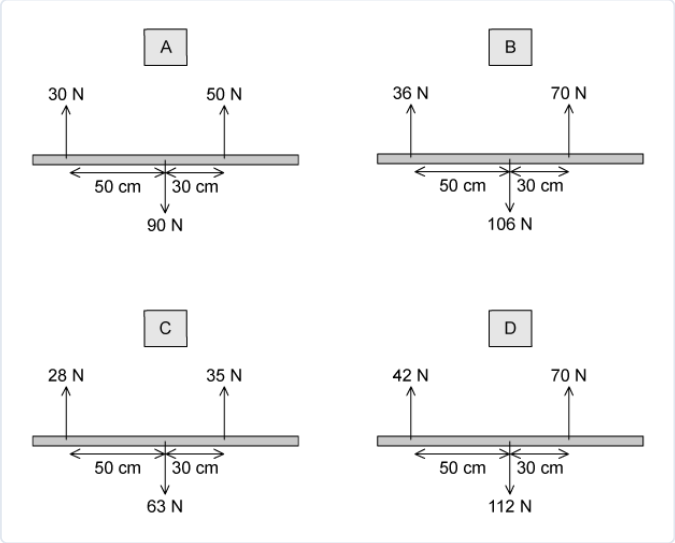
When dealing with systems in equilibrium (e.g. moments, conservation of momentum) you don't need to convert units into SI units, you just need to make sure the units are the same on both sides and you need to check your final answer is reported in the correct units, take this question as an example:



Question 8 Medium [1 mark]

Four beams of the same length each have three forces acting on them.

Which beam has both zero resultant force and zero resultant torque acting?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

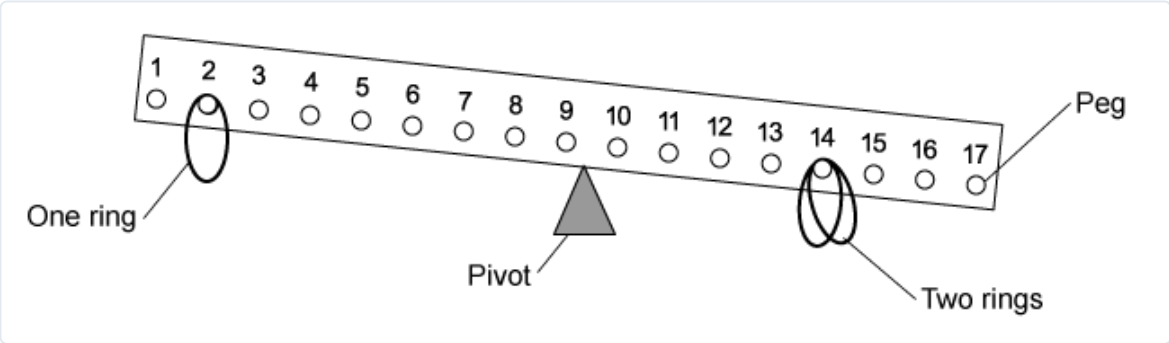
- Torque is the moment of a force, it causes a turning effect
- Consider option **D**
- Upward force = 42 + 70 = 112 N
- Downward force = 112 N
    - Resultant force = 112 – 112 = 0
- Clockwise moment = 42 × 50 = 2100 N cm
- Anticlockwise moment = 70 × 30 = 2100 N cm
    - Resultant moment = 2100 – 2100 = 0
- Both the resultant force and moments are zero, so **D** is correct

A is incorrect because	Upward force = 30 + 50 = 80 N

	Downward force = 90 N • Resultant force = 90 – 80 = 10 N downward Clockwise moment = 30 × 50 = 1500 N cm Anticlockwise moment = 50 × 30 = 1500 N cm • Resultant moment = 1500 – 1500 = **0 N cm**
B is incorrect because	Upward force = 36 + 70 = 106 N Downward force = 106 N • Resultant force = 106 – 106 = 0 Clockwise moment = 36 × 50 = 1800 N cm Anticlockwise moment = 70 × 30 = 2100 N cm • Resultant moment = 2100 – 1800 = **300 N cm**
C is incorrect because	Upward force = 28 + 35 = 63 N Downward force = 63 N • Resultant force = 63 – 63 = 0 Clockwise moment = 28 × 50 = 1400 N cm Anticlockwise moment = 35 × 30 = 1050 N cm • Resultant moment = 1400 – 1050 = **350 N cm**

Question 9 Medium [1 mark]

The diagram shows a child's balancing game.



The wooden rod is uniform and all the rings are of equal mass. Two rings are hung on peg 14 and one on peg 2. The rod is pivoted at peg 9.

On which hook must a fourth ring be hung in order to balance the rod?

A 2

B 3

C 5

D 6

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The weight of one ring is equal to *mg*
- In order for the rod to be in equilibrium :
- Clockwise moments = Anticlockwise moments
  - Clockwise moment = $2 \times 14 - 9mg = 10 \quad mg$
- Let *x* be the position of the hook where the fourth ring will be hung
  - Anticlockwise moment = $9 - 2mg + 9 - xmg$
- For equilibrium:
  - $7 \quad mg + 9 - xmg = 10 \quad mg$
  - $9 - xmg = 3mg$

- $9 - x = 3$
- $x = 9 - 3 = 6$

Question 10 Medium [1 mark]

An object, made from two equal masses joined by a light rod, falls with uniform speed through air. The rod remains horizontal.

Which statement about this system is correct?

A it is not in equilibrium because it is falling steadily

B it is not in equilibrium because it is in motion

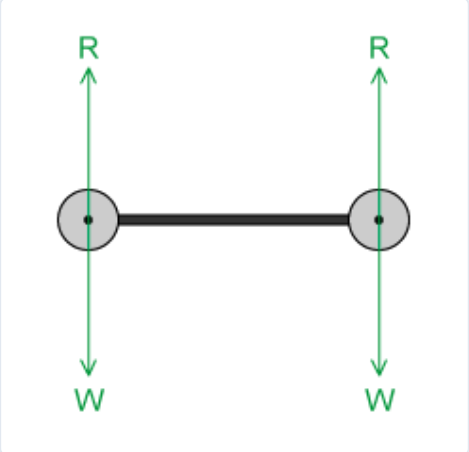
C it is not in equilibrium because there is a resultant torque

D it is in equilibrium because there is no resultant force and no resultant torque

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- For an object to be in equilibrium, it must be experiencing no acceleration. This means that both the net force and the net torque on the object must be zero
- Since the masses are both equal, their weight will be the same and pointing downwards, so the clockwise and anticlockwise moments will be the same, therefore their resultant torque must be zero
- We are told the object moves with uniform speed, or terminal velocity, this means the air resistance on the object will be same as the weight and hence there is no acceleration or resultant force
- Therefore, the object satisfies both conditions for equilibrium

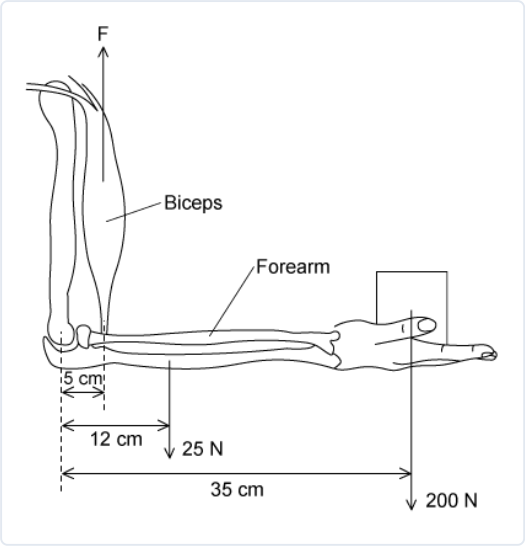


A & B are incorrect because both these statements argue that the object cannot be in equilibrium if the object is moving, however, this is not true, as long as an object is not accelerating then it can be in equilibrium

C is incorrect because a resultant torque could only occur if the forces were unbalanced, however, the object moves with constant speed so the forces must all be balanced

Question 11 Medium [1 mark]

A man holds a 200 N load stationary in his hand. The combined weight of the forearm and hand is 25 N. The forearm is held horizontal, as shown



What is the vertical force *F* needed in the biceps?

A 1340 N

B 1400 N

C 1460 N

D 1520 N

Mark Scheme & Explanation:

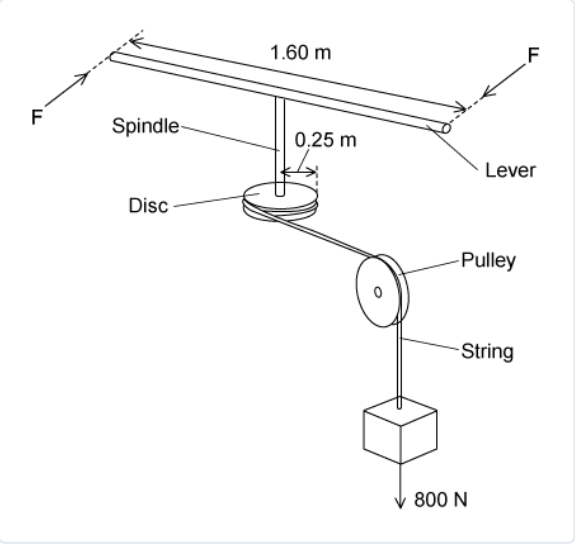
*The correct answer is **C** because:*

- We are told the mass is stationary in this person's hand, therefore, the system is in equilibrium
- For a system in equilibrium :
  - Sum of clockwise moments = Sum of anticlockwise moments
- Clockwise moments = $25 \times 12 + 200 \times 35 = 7300 \quad \text{N} \quad \text{cm}$
- Anticlockwise moments = $5F$
  - $5F = 7300$

- $F = \frac{7300}{5} = 1460 \text{ N}$

Question 12 Medium [1 mark]

A spindle is attached at one end to the centre of a lever of length 1.60 m and at its other end to the centre of a disc of radius 0.25 m. A string is wrapped round the disc, passes over a pulley and is attached to an 800 N weight.



What is the minimum force *F*, applied to each end of the lever, that could lift the weight?

A 62.5 N

B 125 N

C 250 N

D 750 N

Mark Scheme & Explanation:

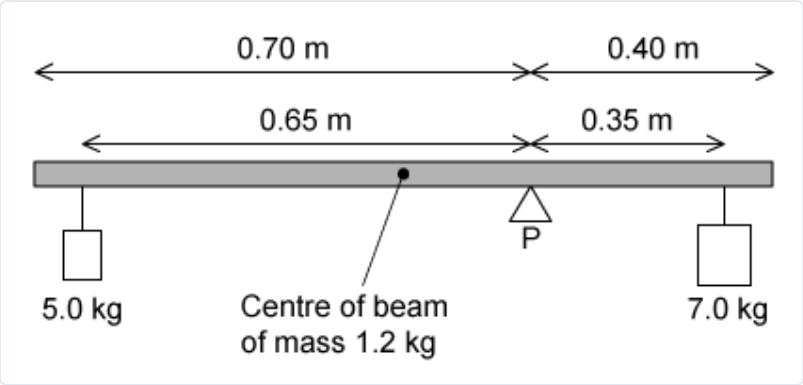
*The correct answer is **B** because:*

- Distance of force *F* from the centre of the spindle $= \frac{1.60}{2} = 0.80 \quad \text{m}$
- Both forces produce clockwise moments about the spindle
  - Clockwise moment $= F \times 0.80 + F \times 0.80 = 1.60 \quad F$
- The 800 N weight acts at a distance 0.25 m from the spindle which causes an anticlockwise moment
  - Anticlockwise moment $= 800 \times 0.25$
- The minimum force that can be applied when the clockwise moments are equal to the anticlockwise moments
  - $F \times 0.80 + F \times 0.80 = 800 \times 0.25$

- $1.60 \quad F = 200$
- $F = \frac{200}{1.6} = 125 \quad \text{N}$

Question 13 Medium [1 mark]

A uniform beam of length 1.10 m and mass 1.2 kg is pivoted at P as shown. The pivot P is 0.40 m from one end. Loads of masses 5.0 kg and 7.0 kg are suspended 0.65 m and 0.35 m from the pivot, respectively, as shown.



What torque must be applied to the beam in order to maintain it in equilibrium?

A 0.62 N m

B 0.98 N m

C 6.1 N m

D 9.6 N m

Mark Scheme & Explanation:

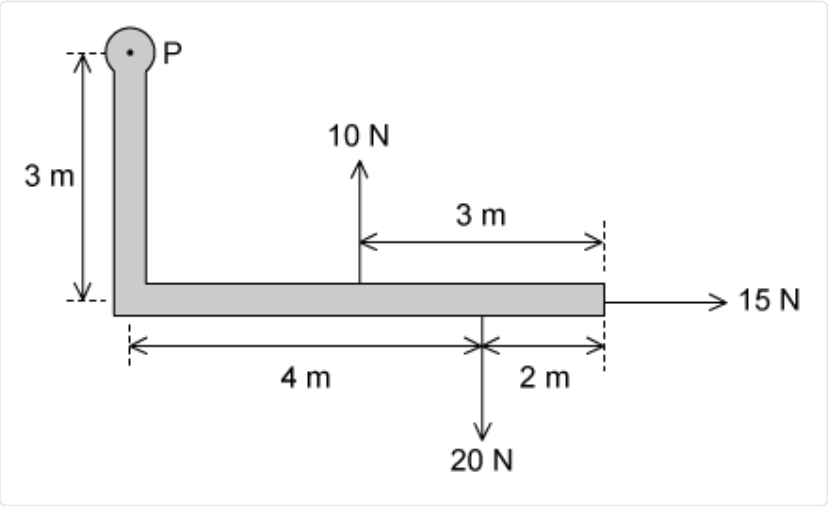
*The correct answer is **D** because:*

- To calculate moments, we will need to use
  - $\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$
- Let the torque be equal to *T*, and assume it acts in a clockwise direction unless it is calculated to have a negative sign (which would tell us it acts in an anticlockwise direction)
- Clockwise moments:
  - $0.35 \times 5.0 \times g + T$
- The beam is uniform, meaning its weight acts at the centre
- The length of the beam is 1.10 m, so the centre mark is at 0.55 m, which is 0.15 m from the pivot
- Anticlockwise moments:
  - $0.15 \times 1.2 \times g + 0.65 \times 5.0 \times g$

- For the system to be in equilibrium, clockwise moment = anticlockwise moment
  - $0.15 \times 1.2 \times g + 0.65 \times 5.0 \times g = 0.35 \times 7.0 \times g + T$
  - $0.18g + 3.25g = 2.45g + T$
  - $T = 0.98g = 9.6 \quad \text{N} \quad \text{m}$
- Therefore, the torque required to keep the beam in equilibrium is 9.6 N m clockwise

Question 14 Medium [1 mark]

A rigid L-shaped lever arm is pivoted at point P.



Three forces act on the lever arm, as shown in the diagram.

What is the magnitude of the resultant moment of these forces about point P?

A 5 N m

B 65 N m

C 95 N m

D 155 N m

Mark Scheme & Explanation:

*The correct answer is **A** because:*

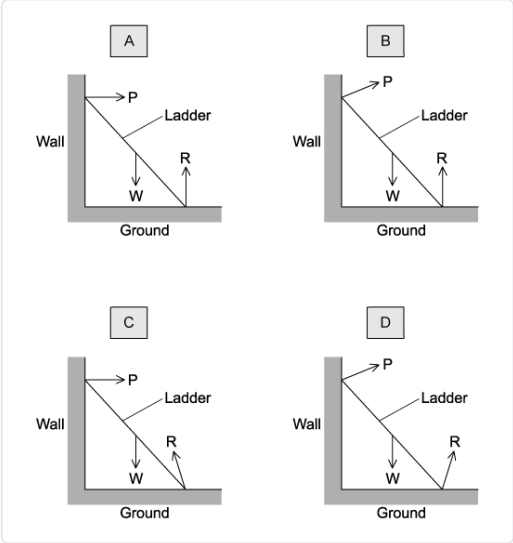
- The force which will cause the lever to rotate clockwise is the 20 N force
- The forces which will cause the lever to rotate anticlockwise are the 10 N and 15 N forces
- Perpendicular distance of 10 N force : (4 + 2) – 3 = 3 m
- Perpendicular distance of 15 N force : 3 m
- Perpendicular distance of 20 N force : 4 m
- Magnitude of resultant moment = Clockwise moment – Anticlockwise moment

- Magnitude of resultant moment $= 20 \times 4 - 10 \times 3 - 15 \times 3 = 5 \quad \mathrm{N} \quad \mathrm{m}$

Question 15 Medium [1 mark]

A ladder is positioned on icy ground and is leant against a rough wall. At the instant of release, it begins to slide.

Which diagram correctly shows the directions of the forces *P*, *W* and *R* acting on the ladder as it slides?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

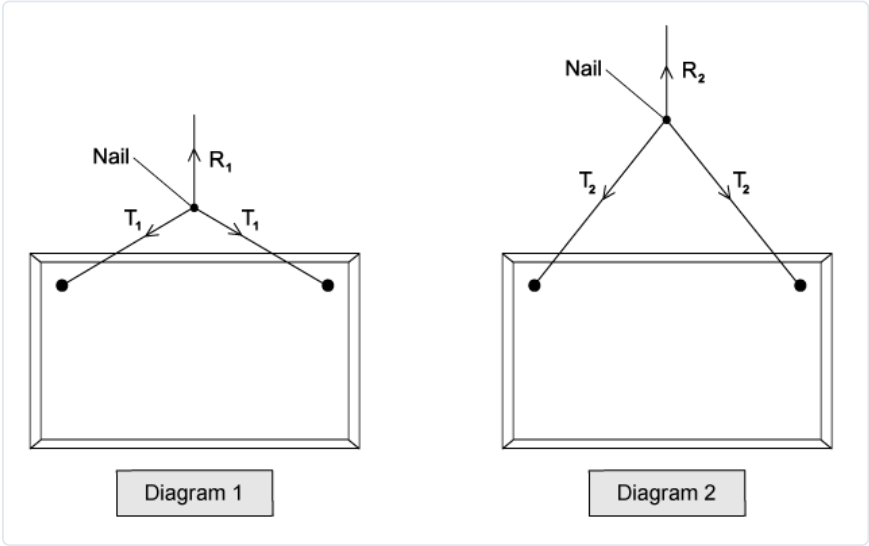
- The weight *W* always acts downwards
- The reaction force *R* acts in the opposite direction to the weight, upwards
- The ground is said to be icy which means it is **frictionless**, so there are no additional forces acting or opposing the motion
- Before the ladder begins to slide, there is a horizontal force *P* acting to the right where the ladder comes into contact with the rough wall because the ladder exerts a force to the left on the wall
- As the ladder begins to slide, the point of contact of the ladder with the wall begins to move down the wall
- We are told the wall is rough, meaning friction will oppose the motion of the ladder as it begins to move
- So, at the point of contact, *P* is now acting to the right along with the upwards frictional force, which leads to a resultant force as shown in diagram B

A & C are incorrect because these diagrams show the force *P* as horizontal on the ladder as before it begins to slide there is initially no friction force acting, however the question asks about the forces on the ladder **as** it begins to slide

C & D are incorrect because these diagrams show that there is a frictional force acting on the ground causing the resultant force of *R* to change direction, but this is incorrect since the ground is icy, meaning it is frictionless

Question 16 Hard [1 mark]

The diagrams show two ways of hanging the same picture.



In both cases, a string is attached to the same points on the picture and looped symmetrically over a nail in a wall. The forces shown are those that act on the nail.

In diagram 1, the string loop is shorter than in diagram 2.

Which information about the magnitude of the forces is correct?

A	$R_1 = R_2$	$T_1 < T_2$
B	$R_1 = R_2$	$T_1 > T_2$
C	$R_1 > R_2$	$T_1 < T_2$
D	$R_1 < R_2$	$T_1 = T_2$

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In each case, the sum of the vertical components of the two tensions equals the same weight W
- So, the reaction at the nail has the same magnitude in both cases:
    - $R_1 = R_2$
- To determine which is the larger of T_1 or T_2, we must consider the angles between the strings and the vertical, let's call these θ_1 and θ_2
- For each diagram, the sum of the vertical downward components of the tensions should be equal to the weight which is also equal to R
    - $W = 2T_1 \cos \theta_1$
    - $W = 2T_2 \cos \theta_2$
    - $2T_1 \cos \theta_1 = 2T_2 \cos \theta_2$
- It is clear from the diagrams that the angle θ_1 is larger than θ_2
- From the graph of cosine, we know that as the angle θ increases, the value of cos θ decreases
- Since $\theta_1 > \theta_2$, cos θ_1 is smaller than cos θ_2, so for the statement to hold true, we can conclude:
    - $T_1 > T_2$

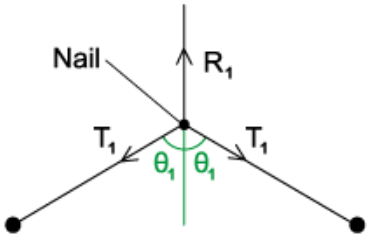


Diagram 1

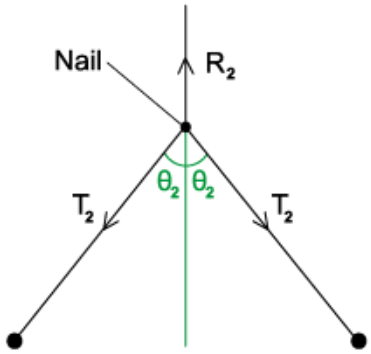


Diagram 2

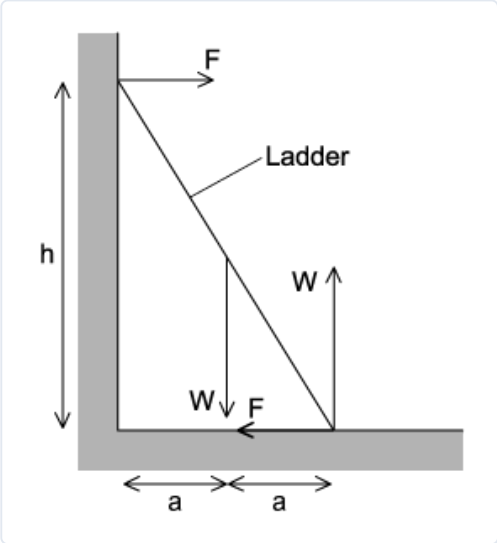
A is incorrect because this implies that $\theta_1 < \theta_2$, which is incorrect

C & **D** are incorrect because R is equal to the weight of the picture so both of these are incorrect

Question 17 Hard [1 mark]

A uniform ladder rests against a smooth vertical wall, as shown in the diagram. The bottom of the ladder rests on rough ground where friction is not negligible. The top of the ladder is a vertical height h up the wall, and the foot of the ladder is a horizontal distance $2a$ from the wall.

The weight W of the ladder acts at its centre. At the foot of the ladder, the ground exerts a vertical reaction R and a horizontal frictional force F.



Which equation represents the moments when the ladder is in equilibrium?

A $Wa + Fh = 2Wa$

B $Fa + Wa = Fh$

C $Wa + 2Wa = Fh$

D $Wa - 2Wa = 2Fh$

Mark Scheme & Explanation:

*The correct answer is **A** because:*

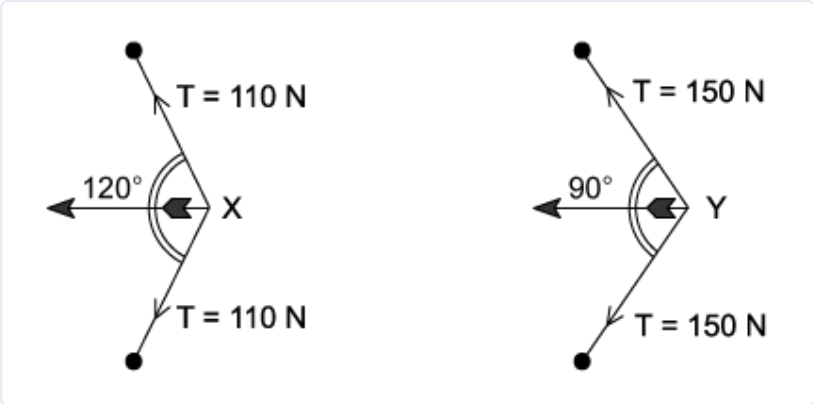
- The moment of a force is equal to
  - $\text{moment} = \text{force} \times \text{perpendicular distance from pivot}$
- Taking moments about the top of the ladder
- **Clockwise moments:**

- The horizontal frictional force F acting at a perpendicular distance h produces a clockwise moment = $F \times h$
- The weight W of the ladder acting at a perpendicular distance a produces a clockwise moment = $W \times a$

- **Anticlockwise moments:**
  - The vertical reaction force $R = W$ acting at a perpendicular distance $2a$ produces an anticlockwise moment = $W \times 2a$
- For equilibrium: Sum of clockwise moments = Sum of anticlockwise moments
  - $Wa + Fh = 2Wa$

Question 18 Hard [1 mark]

An archer draws his bowstring back to position X. The bowstring and arrow are shown. The tension *T* in the string is also shown. Then he draws the bowstring back further to position Y.



The archer finds the resultant force on the arrow is greater when the arrow is released from position Y.

What is the increase in force?

A 51 N

B 55 N

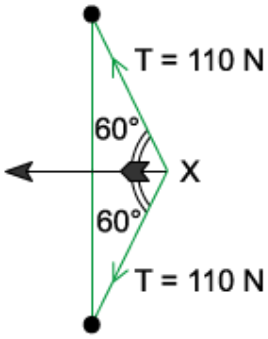
C 102 N

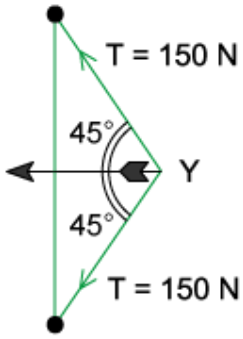
D 110 N

Mark Scheme & Explanation:

The correct answer is ***C*** *because:*

- Hopefully you will have quickly guessed that this problem will require trigonometry to solve
- In both cases, the bowstring is in equilibrium, so we can work out the resultant force in each bow using:
  - $F = T \cos \frac{\theta}{2} + T \cos \frac{\theta}{2} = 2T \cos \frac{\theta}{2}$
- Where *T* is the tension in the string and *θ* is the angle between the strings
- The 2 *T* is because there is tension on both halves of the string and the ½ *θ* is because you resolve in line with the direction that the archer is pulling, which is half the angle between the halves of the bow
  - Resultant force in X = $2T \cos \frac{\theta}{2} = 2 \times 110 \cos \frac{120}{2} = 220 \cos 60$
  - Resultant force in Y = $2T \cos \frac{\theta}{2} = 2 \times 150 \cos \frac{90}{2} = 300 \cos 45$
  - Increase in force = $300 \cos 45 - 220 \cos 60 = 102$ N

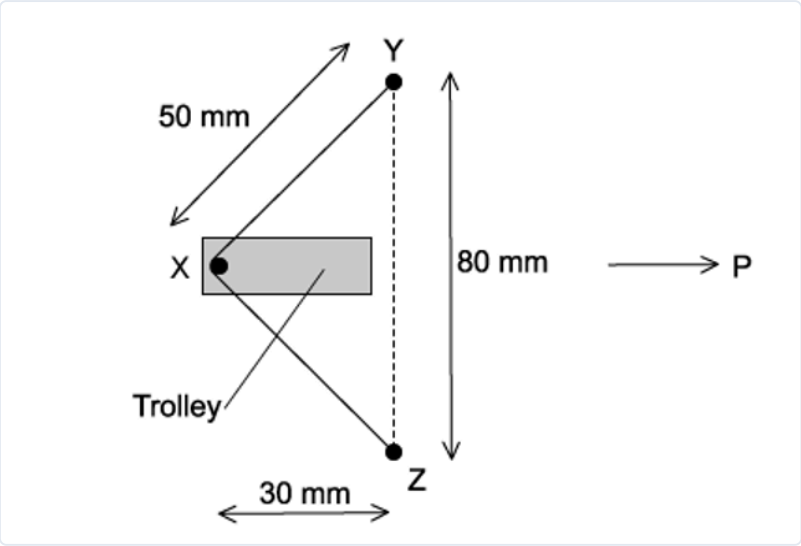




Exam Tip: if there is more than one force triangle in a question, always consider each one separately, many students consider the two strings as one and end up with the wrong answer, so be careful!

Question 19 Hard [1 mark]

The diagram shows two fixed pins, Y and Z. A length of elastic is stretched between Y and Z and around pin X, which is attached to a trolley.



X is at the centre of the elastic and the trolley is to be propelled in the direction P at right angles to YZ. The tension in the elastic is 4 N.

What is the force accelerating the trolley in the direction P when the trolley is released?

A 2.4 N

B 3.2 N

C 4.8 N

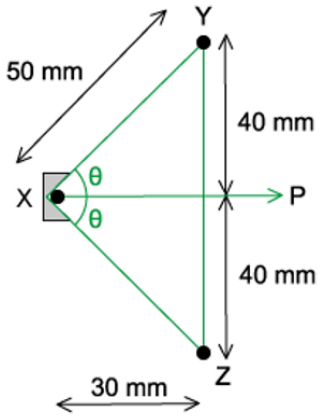
D 6.4 N

Mark Scheme & Explanation:

*The correct answer is **C** because:*

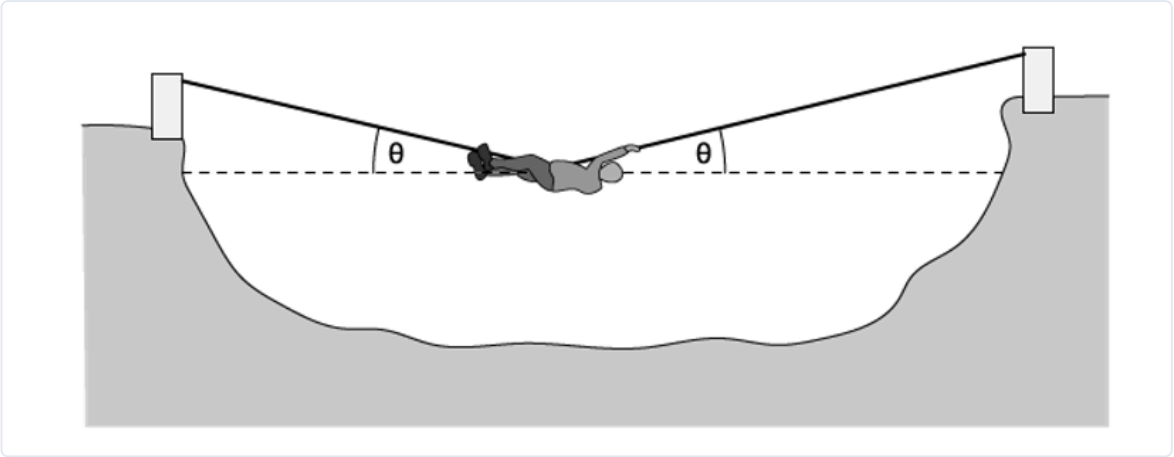
- The component of the tension in each string along the direction XP is $T \cos \theta$
  - Where T is the tension in the string and θ is the angle between segment XY and axis XP
- The elastic string is in equilibrium, so we can work out the resultant force using:
  - $F = T \cos \theta + T \cos \theta$
  - $F = 2T \cos \theta$
- The $2T$ arises because there is tension on both halves of the string

- The quantity cos θ can be found using
  - $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{30}{50} = \frac{3}{5}$
- Since T = 4 N, the resultant force is equal to
  - $F = 2 \times 4 \times \frac{3}{5}$
  - $F = 4.8 \text{ N}$



Question 20 Hard [1 mark]

The diagram shows a rope bridge that a student makes on an adventure training course. The student has a weight *W*.



Which formula gives the tension *T* in the rope?

A $T = \frac{W}{2 \cos \theta}$

B $T = \frac{W}{2 \sin \theta}$

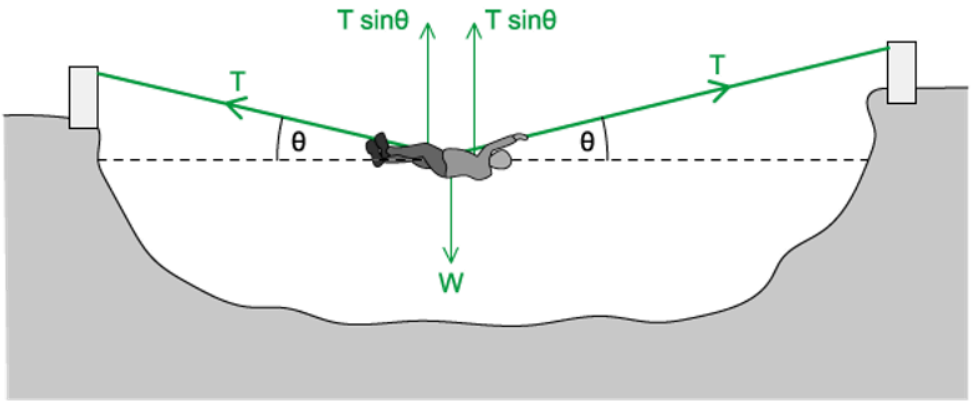
C $T = \frac{W}{\cos \theta}$

D $T = \frac{W}{\sin \theta}$

Mark Scheme & Explanation:

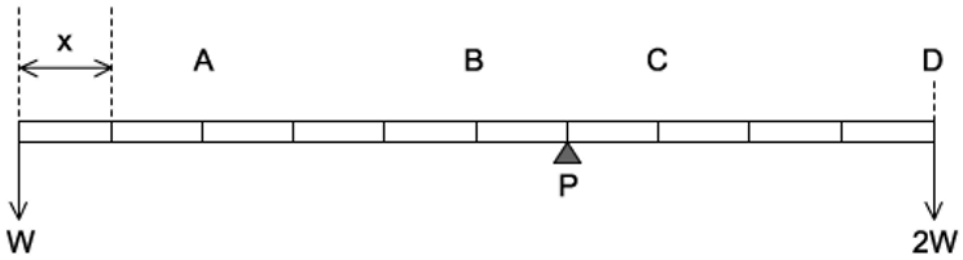
*The correct answer is **B** because:*

- There is a tension *T* in the rope on each side of the student
- The sum of vertical components of tension should be equal to the weight *W*
- The vertical component of tension is equal to
  - $T \sin \theta$
- The student is in equilibrium and the vertical forces are shown as in the diagram below, so:
  - $T \sin \theta + T \sin \theta = W$
  - $2T \sin \theta = W$
  - $T = \frac{W}{2 \sin \theta}$


T sinθ
T sinθ
T
T
θ
θ
W

Question 21 Hard [1 mark]

A uniform beam is pivoted at P as shown. Weights of *W* and 2 *W* are attached to its ends. The beam is marked at *x* intervals. The weight of the beam is 10 *W*



At which point should a 2 *W* weight be attached to achieve equilibrium?

A

B

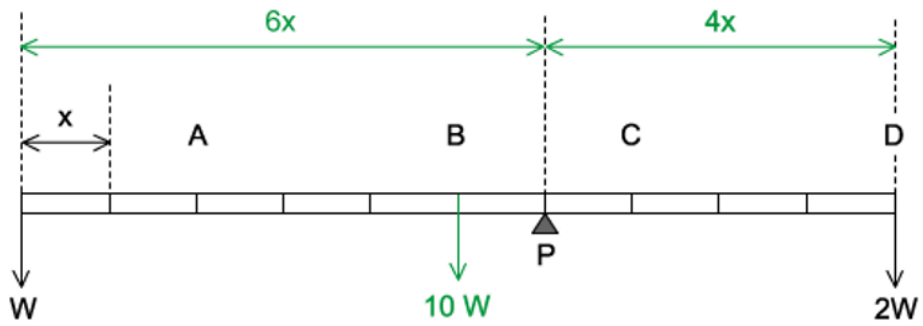
C

D

Mark Scheme & Explanation:

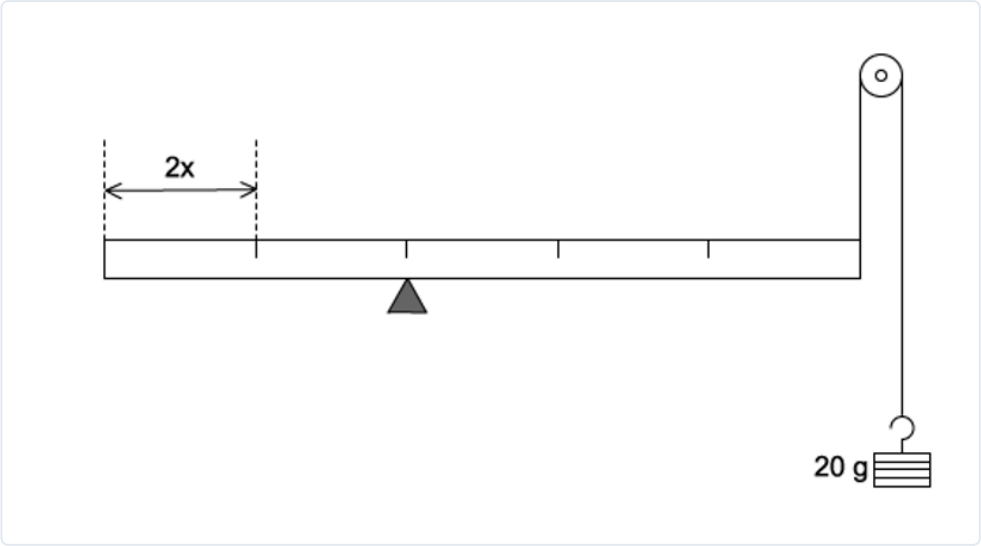
*The correct answer is **D** because:*

- Since the beam is uniform, its weight acts at the centre which is a distance *x* to the left of the pivot
- For equilibrium : Sum of clockwise moments = Sum of anticlockwise moments
- Clockwise moments $= 2W \times 4x = 8Wx$
- Anticlockwise moments $= W \times 6x + 10W \times x = 6Wx + 10Wx = 16Wx$
- Since the anticlockwise moment is greater, the 2*W* weight should contribute to the sum of the clockwise moments, i.e. to the right of the pivot
- The 2 *W* weight acts at a distance *d* from the pivot
- Sum of clockwise moments = Sum of anticlockwise moments
  - $8Wx + 2Wd = 16Wx$
  - $2Wd = 8Wx$
  - $d = 4x$
- A distance of 4*x* to the right corresponds to option D


6x
4x
x
A
B
C
D
P
W
10 W
2W

Question 22 Hard [1 mark]

A uniform metre rule of mass 100 g is supported by a pivot and a string as shown in the diagram. The beam is marked at intervals of $2x$. The string passes round a frictionless pulley and carries a mass of 20 g.



At which mark on the rule must a 50 g mass be suspended so that the rule balances?

A $0.4x$

B $3.6x$

C $4.4x$

D $6.4x$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

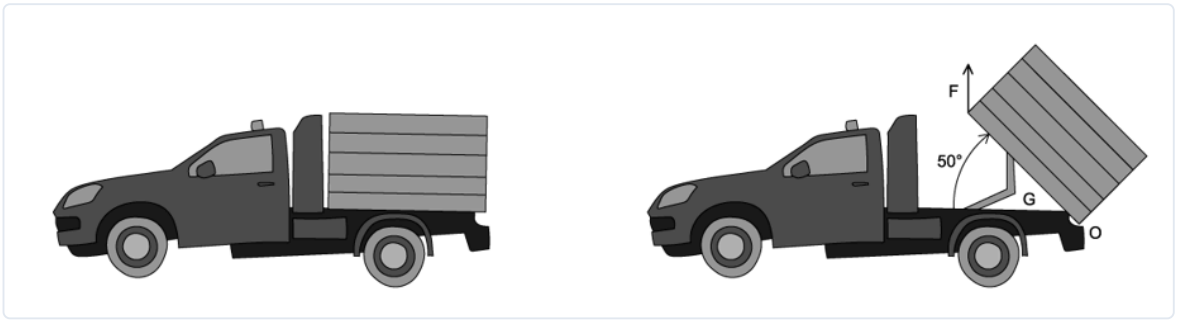
- Since the metre rule is uniform, its weight acts at its centre, which is 5*x*
- The weight of the ruler causes a clockwise moment at a distance 5*x* – 4*x* = *x* from the pivot
  - Clockwise moment due to the weight of the ruler = 100 *gx*
- Where *g* is the acceleration due to gravity
- The weight of the 20 g mass causes an anticlockwise moment at a distance of 10*x* – 4*x* = 6*x* from the pivot
  - Anticlockwise moment due to 20 g mass = $20g \times 6x = 120gx$
- For equilibrium, the clockwise moment should equal the anticlockwise moment

- Since the anticlockwise moment is greater, the 50 g mass should contribute to the sum of the clockwise moments, i.e. to the right of the pivot
- Let's say the 50 g mass acts at a distance *d* from the pivot
- Sum of clockwise moments = Sum of anticlockwise moments
  - $100gx + 50gd = 120gx$
  - $50gd = 20gx$
  - $d = \frac{20x}{50} = 0.4x$
- Therefore, the 50g mass should be hung at
  - $4x + 0.4x = 4.4x$

Question 23 Hard [1 mark]

A trailer of weight 30 kN is attached to a truck. The trailer is raised until it is held at 50˚ to the horizontal by a vertical force *F* as shown in the diagram.

The trailer is 2.7 m in length and the position of its centre of gravity G is a third of the length from the hinge at O.



What is the vertical force *F* applied to the end of the trailer?

A 10 kN

B 15 kN

C 30 kN

D 60 kN

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- So far, we have just looked at situations where the object is uniform so the centre of mass acts through the centre of the object
- We are told the position of its centre of gravity G is a third of the length from the hinge at O so the weight acts at a distance of $\frac{2.7}{3} = 0.9 \text{ m}$ from O
- For equilibrium, the clockwise moment should equal the anticlockwise moment
- Let's consider moments around the pivot O
- The vertical force *F* acts at a perpendicular distance of 2.7 cos 50
- The weight of the trailer acts at a perpendicular distance of 0.9 cos 50
- The vertical force *F* provides a clockwise moment and the weight of the trailer provides an anticlockwise moment
    - Clockwise moment due to force *F* = $F \times 2.7 \cos 50$

- Anticlockwise moment due to the weight = $30000 \times 0.9 \cos 50$

- Sum of clockwise moments = Sum of anticlockwise moments
  - $F \times 2.7 \cos 50 = 30000 \times 0.9 \cos 50$
  - $F = \frac{30000 \times 0.9 \cos 50}{2.7 \cos 50} = 10000 \text{ N}$

- The vertical force is therefore equal to 10 000 N or 10 kN

